SAIFUDIN NASIR

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Recent Informatics Engineering graduate from Gunadarma University with specialized expertise in Machine Learning and Full-stack Web Development. Currently contributing to a strategic university research project developing Hybrid ML and Generative AI models for national food security. Proven ability to build end-to-end applications, from implementing Random Forest algorithms for environmental prediction to developing interactive web dashboards using FastAPI and Python.

WORK & RESEARCH EXPERIENCE

Universitas Gunadarma | ML Programmer & Full-stack Developer Agustus - Desember 2025

- Developing a Hybrid ML and Generative AI model to predict national rice availability for food security planning.
- Building and optimizing Backend APIs using FastAPI (Python) for efficient model inference and data processing.
- Implementing **Frontend** interfaces by converting UI designs into functional, responsive web dashboards for real-time data visualization.

Binus University | IT Support Agustus - November 2020

- Performed diagnostics, assembly, and repair for PCs, laptops, and peripherals.
- Maintained network connectivity (internet/intranet) and resolved connection issues.

Provided helpdesk support by responding to technical requests and managing software/OS installations.

TECHNICAL PROJECTS

Rice Availability Dashboard (Hybrid ML & GenAI) | *University Research Project*

- Architected a robust Backend using FastAPI to integrate predictive models with a real-time web system.
- Developed a comprehensive Audit Logging system and secure middleware to prevent vulnerabilities like Clickjacking and MIME sniffing.
- Implemented advanced security measures, including JWT Authentication (Access & Refresh Tokens), Rate Limiting, and Role-Based Access Control (RBAC).
- Implemented a Hybrid model combining traditional Machine Learning with Generative AI for deep analytical insights.

Stock Market Forecasting Application | *Independent Project*

- Engineered a predictive application leveraging XGBoost and LSTM algorithms for financial time-series analysis.
- Automated data extraction using the yfinance API and integrated multi-source sentiment analysis from news outlets and social media platforms.

Web-Based Mangrove Carbon & Biomass Prediction | *Final Project*

- Designed a web-based platform for mapping and predicting environmental data using Random Forest.
- Managed the full development lifecycle, from data preprocessing in Python to final web deployment.

IoT Indoor Air Quality Monitoring Tool | *Scientific Research*

- Constructed an integrated hardware-software system using **NodeMCU ESP8266** for real-time monitoring.

EDUCATIONS

UNIVERSITAS GUNADARMA 2021 – 2025

Bachelor's Degree in Informatics – GPA 3.59 / 4.00

(*Awaiting official graduation letter / SKL*)

- **Concentrations**: Machine Learning, Data Science, Database Management, Web Development, Data Quality & Governance, Project & Event Management.
- **Scientific Research**: Design and Implementation of an Indoor Air Quality Monitoring Tool using NodeMCU ESP8266
- **Final Project**: Mapping and Prediction of Mangrove Carbon and Biomass Using Web-Based Random Forest Machine Learning

SMKN 29 JAKARTA 2018 – 2021

Industrial Electronics Engineering

SKILLS

- **Technical:** Python (FastAPI, Scikit-learn, TensorFlow, XGBoost), Backend Security (JWT, RBAC, Rate Limiting), PostgreSQL, SQL, PHP, Java.
- **AI/DS:** Machine Learning, Generative AI, Data Analysis, Feature Engineering.
- **Web & Tools:** Audit Logging, Secure API Development, IoT (NodeMCU), Windows CMD, Microsoft Office, Canva, Figma
- **Languages:** Bahasa Indonesia (Native), English (Basic).

TRAININGS AND CERTIFICATIONS

- **Junior Web Programmer** – BNSP Certified.
- **Networking & Cisco Fundamentals** – Gunadarma University.
- **Mikrotik Fundamentals (Beginner & Intermediate)** – Gunadarma University.